

Timothy O. Trujillo

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Visiting Professor, Sam Houston State University

<https://trujillo1234.github.io/> | Updated October 6, 2026

Research interests

Mathematical logic, set theory, ultrafilters, topological Ramsey spaces, nonstandard models of arithmetic, and mathematical foundations of collective choice. Current proposed research concerns infinite ethics and universal consequentialist systems with compatible aggregation and evaluation.

Academic appointments

Sam Houston State University Visiting Professor, September 1, 2026-present. Working on research projects.

Sam Houston State University, Mathematics and Statistics Associate Professor of Mathematics, September 2023-August 2026. Promoted with tenure effective September 1, 2023. Assistant Professor of Mathematics, 2017-2023.

Colorado School of Mines Postdoctoral Teaching Scholar in Mathematics, 2015-2017.

Texas State University Lecturer in Mathematics, 2014-2015.

University of Denver Instructor and Graduate Teaching Assistant in Mathematics, 2008-2014.

New Mexico Institute of Mining and Technology Instructor in Mathematics, 2006-2008.

Education

Ph.D. in Mathematics, University of Denver, 2014. Advisor: Natasha Dobrinen. Dissertation: Topological Ramsey spaces, associated ultrafilters and their applications to the Tukey theory of ultrafilters and Dedekind cuts of nonstandard arithmetic.

M.S. in Mathematics, New Mexico Institute of Mining and Technology, 2008. Specialization in analysis. Advisor: Bixiang Wang. Thesis: Attractors for nonlinear reaction-diffusion equations in unbounded domains.

B.S. in Mathematics, New Mexico Institute of Mining and Technology, 2006. Minor in education.

Selected publications

Timothy Trujillo. Parametrizing by the Ellentuck Space. *Fundamenta Mathematicae* 255 (2021). Publisher record

Timothy Trujillo. Condorcet's Paradox and Ultrafilters. *The American Mathematical Monthly* 127(2) (2020), 169-174. DOI: 10.1080/00029890.2020.1680200

Timothy Trujillo, Connor Mattes, J. Menard, and Zachary Chaney. Triangular Ramsey numbers. *INTEGERS* 19 (2019). Article PDF

Natasha Dobrinen, Jose Mijares, and Timothy Trujillo. Topological Ramsey spaces from Fraisse classes, Ramsey-classification theorems, and initial structures in the Tukey types of p-points. *Archive for Mathematical Logic* (2017). DOI: 10.1007/s00153-017-0540-0

Timothy Trujillo. Selective but not Ramsey. *Topology and its Applications* 202 (2016). DOI: 10.1016/j.topol.2015.12.088

Timothy Trujillo. Ramsey for R_1 ultrafilter mappings and their Dedekind cuts. *Mathematical Logic Quarterly* 61 (2015). DOI: 10.1002/malq.201300078

Timothy Trujillo and Bixiang Wang. Continuity of strong solutions of the reaction-diffusion equation in initial data. *Nonlinear Analysis: Theory, Methods & Applications* 69 (2008). DOI: 10.1016/j.na.2007.08.032

Research support and current project

SHSU Enhanced Research Grant, Infinite ethics: a non-Archimedean approach. The grant-supported work led to Condorcet's Paradox and Ultrafilters. The current proposal, Infinite Ethics: Toward a Universal Consequentialist System, develops the broader program of context-sensitive composition, non-Archimedean evaluation, and coherent extension of partial systems. Universal existence and extension theorems are research objectives.

Selected mentoring, teaching, and service

Directed undergraduate and graduate studies in mathematical logic, set theory, and social choice; supervised student work leading to conference presentations and publications. Co-advised Colorado School of Mines Putnam teams in 2015 and 2016. Taught undergraduate mathematics and graduate mathematical logic at SHSU, with course development in proof-based mathematics and mastery-oriented assessment.

Service at SHSU included chairing a hiring committee for two tenure-track mathematics positions, department policy work, and co-organizing the logic seminar. Project NExT Fellow, Mathematical Association of America, 2017; Texas Section NeXT Fellow, 2017-2018.